EXERCISES

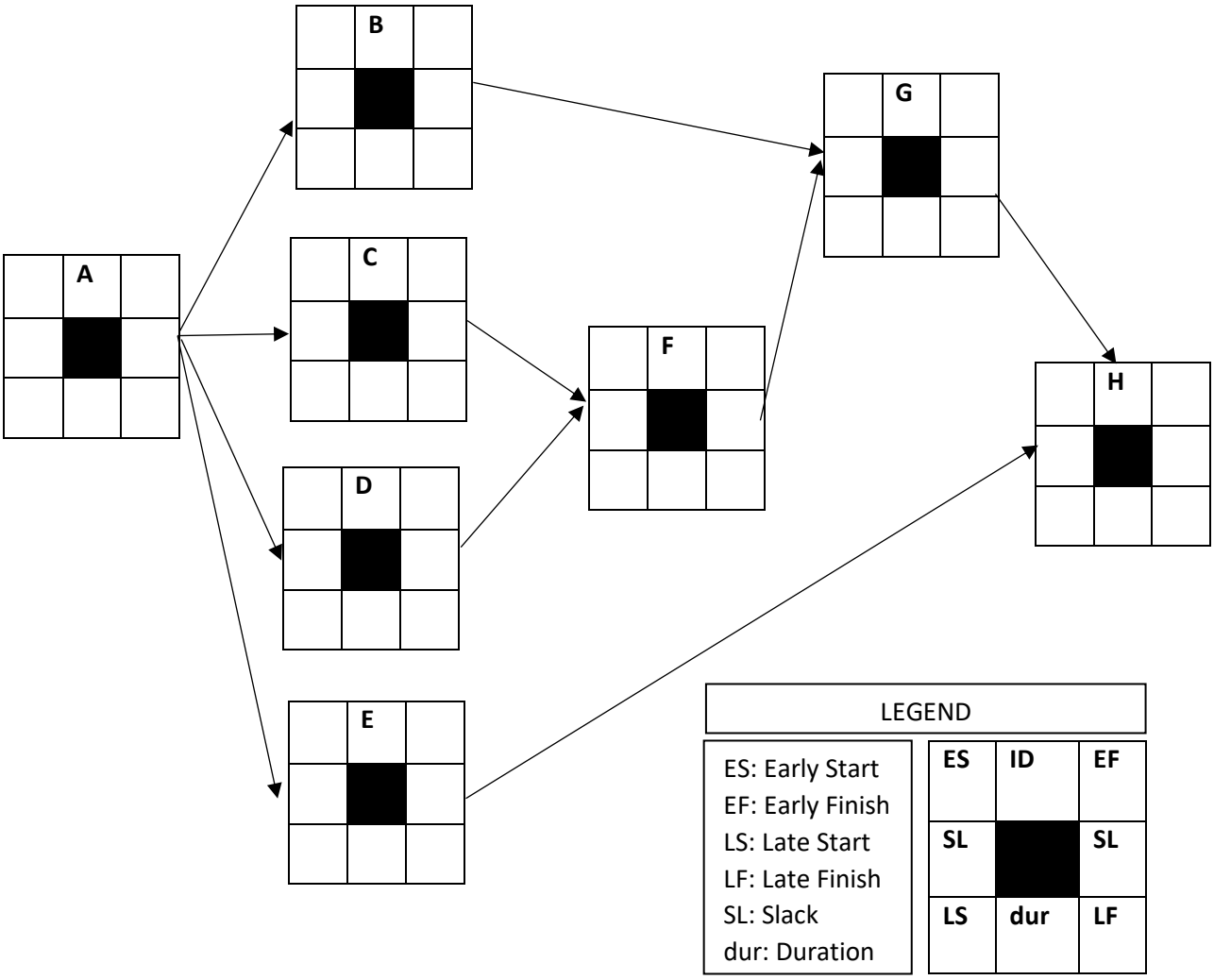
1.Air Control Company started a Project that consist of eight activities respectively, A, B, C, D, E, F, G and H. The duration of eache activity is given below.

Activity	Duration
A	**2**
B	**15**
C	**10**
D	**13**
E	**18**
F	**15**
G	**10**
H	**5**

The Project Manager drew the network diagram that is demonstrated below. Please fill the ES, EF, LS, LF,SL, and dur for each activity, find the critical path and total duration of the Project.

Critical path:_____

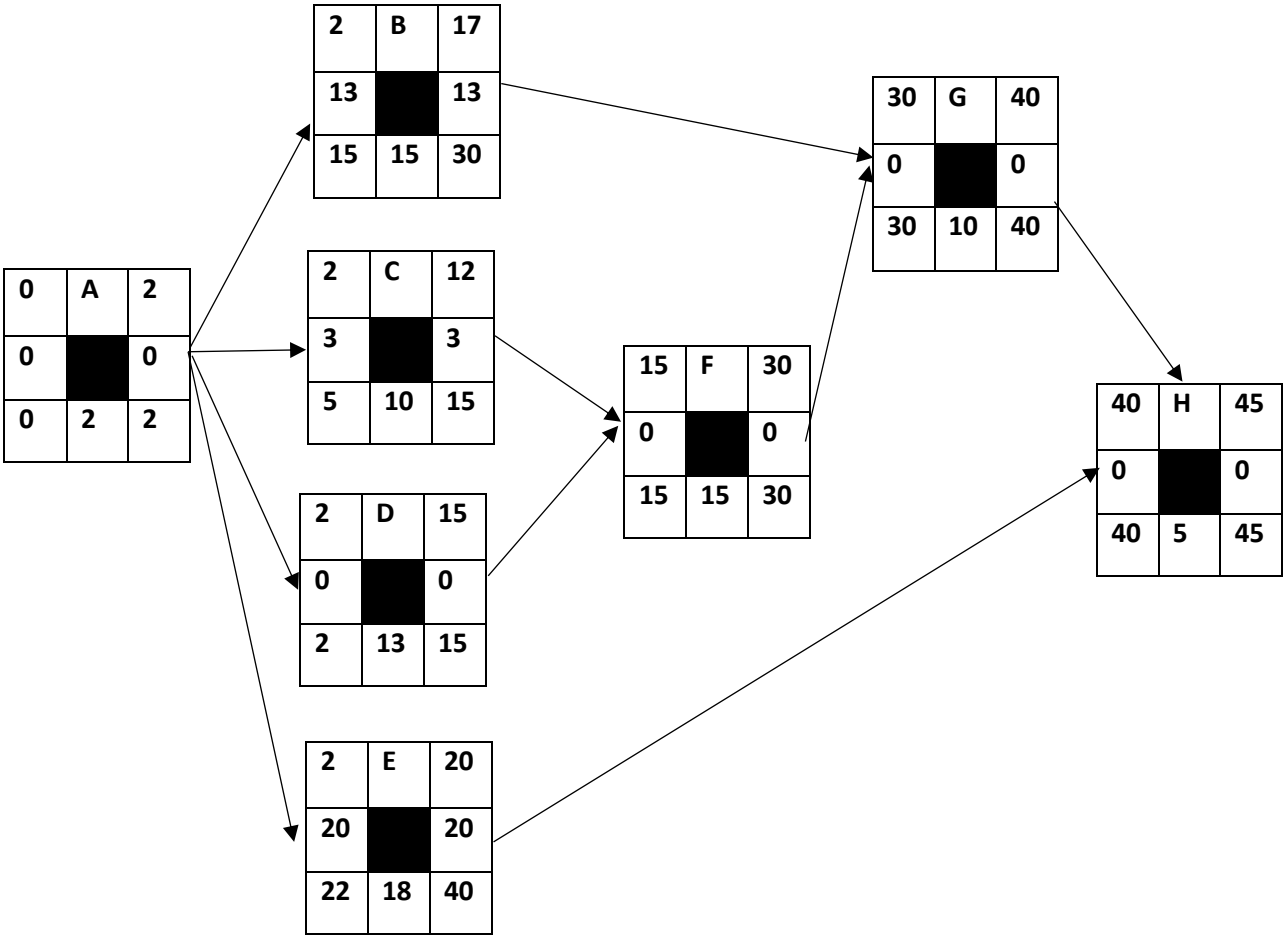
Total duration of the project: _____



SOLUTION:

Critical path:_____A,D,F,G,H_____

Total duration of the project: ___45____



2. Orange Spa decided to develop a new product and hired a project manager to initiate the project. The project manager identified five activities coded as; A, B, C, D and F. The gantt chart diagram of the project that also demonstrates weekly cost of each activity is given below.

Activities	Cost	1	2	3	4	5	6	7	8	9	10	11	12
A	40	10	10	10	10								
B	32	8	4	8	4	8							
C	48					12	12	12	12				
D	18						6	2	2	2	6		
E	28						8	8	12				
F	40											20	20
Total	206	18	14	18	14	20	26	22	26	2	6	20	20
Cumulative		18	32	50	64	84	110	132	158	160	166	186	206

Based on the information given;

i. Please complete the status report of 8th week at the table below.

Activities	% Complete	BCWP	ACWP	BCWS	CV	SV
A	100%		35			
B	100%		24			
C	100%		32			
D	33%		20			
E	100%		20			
Cumulative	Totals		131			

ii. Please calculate the performance indicators. (SPI, CPI) and time based schedule variance (SVtime).

iii. Please calculate the EAC, SAC for the structral problems.

SOLUTION:

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Activities	% Complete	BCWP	ACWP	BCWS	CV	SV
A	100%	40	35	40	5	0
B	100%	32	24	32	8	0
C	100%	48	32	48	16	0
D	33%	6	20	10	-14	-4
E	100%	28	20	28	8	0
Cumulative	Totals	154	131	158	23	-4

CV = BCWP – ACWP
SV = BCWP – BCWS

ii.
CPI = BCWP / ACWP = 154/131 = 1.17
SPI= BCWP / BCWS = 154/158 = 0.97

Activities	Cost	1	2	3	4	5	6	7	8	9	10	11	12
A	40	10	10	10	10								
B	32	8	4	8	4	8							
C	48					12	12	12	12				
D	18						6	2	2	2	6		
E	28						8	8	12				
F	40											20	20
Total	206	18	14	18	14	20	26	22	26	2	6	20	20
Cumulative		18	32	50	64	84	110	132	158	160	166	186	206

This situation, we were already given the weekly costs;
BCWSperiod = Cost of WP / Duration of WP = 12 + 2 + 12 = 26

SVtime = SV / BCWSperiod = -4/26 = -0.15

iii. For structural problems;

EAC= BCWSfinal / CPI = 206/ 1.17 = 176.06
SAC= T final / SPI = 12/ 0.97 = 12.37